The Coherence Barcode: A Template-Matching Diagnostic for Ambient-Noise Phase-Velocity Estimation

FastMSPEC project design document

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Abstract

This document formalizes a diagnostic and phase-velocity-picking method developed for the FastMSPEC project: representing a coherence (noise correlation function, NCF) spectrum's zero-crossings, local maxima, and local minima as a typed sequence of frequency-tagged events — a "barcode" — and judging spectral-estimation techniques (single-taper vs. multitaper methods such as FastMspec) by how well their barcode matches a library of physically-motivated templates, scanned across a plausible range of Love-wave phase velocities. The method is grounded in, but distinct from, three lines of prior work: Ekström et al.'s (2009) zero-crossing phase-velocity construction, Hawkins and Sambridge's (2019) extension of that construction to peaks and troughs via an analytical Bessel-function result, and Xue and Olugboji's (2025) AkiNet architecture, whose bounded-corridor reference-curve design independently converges on the same underlying principle used here. This document is intended as a standalone reference for the method's design, independent of the conversation history that produced it, so that the associated notebook implementation can be revisited, audited, or extended without re-deriving the reasoning from scratch.

Contents

1 Motivation

Ambient-noise interferometry estimates the coherence, or noise correlation function (NCF), between two seismic stations, whose real part is predicted by Aki's (1957) spectral formulation to follow a Bessel-function form. For Rayleigh waves (and, as a common simplifying approximation also used throughout this project, for Love waves too — see Section 4.3),

$$\rho(f, r) \approx J_0\left(\frac{2\pi f r}{c(f)}\right), \tag{1}$$

where $\rho(f, r)$ is the real part of the coherence spectrum, r is the interstation distance, f is frequency, and $c(f)$ is the (generally frequency-dependent) phase velocity.

Extracting $c(f)$ from an observed, noisy $\rho(f, r)$ is a nonlinear inverse problem. A classical approach (Ekström et al., 2009) identifies zero-crossings of the observed spectrum and matches them against the known zero-crossing structure of Equation 1 to construct trial dispersion curves. The FastMSPEC project (this repository) has, over the course of its development, repeatedly found that *how well an estimator's coherence spectrum supports this kind of feature extraction* is itself informative about the estimator's quality — not just its variance, but whether its zero-crossings (and, as developed here, its turning points) sit where a physically plausible earth would put them.

Prior diagnostics built in this project (documented in `notebooks/03_fastmspec_ application.ipynb`, Sections 1c–1f) established two separate findings: (1) single-taper spectral estimates have much higher variance than multitaper (FastMspec) estimates, and (2) smoothing single-taper's estimate to match FastMspec's variance does not recover FastMspec's resolution — smoothed single-taper's zero-crossings still drift substantially, and the amount of drift can be predicted from a resolution-bandwidth criterion, $2W \lesssim \Delta f_{\text{zero}}$, where $\Delta f_{\text{zero}} = c/(2r)$ is the natural zero-crossing spacing implied by the Aki–Bessel relationship. Both findings, however, only checked *internal self-consistency* of a candidate spectrum against itself (e.g., how much do a technique's own crossings drift under smoothing, or how tightly clustered is its own implied zero-crossing spacing). Neither compared a candidate directly against an actual physical model of what a real earth's coherence spectrum should look like.

This document formalizes that missing comparison: the **coherence barcode**.

2 The Barcode

Definition 1 (Barcode)**.** *For any coherence curve $\gamma(f)$ — real (estimated from data) or theoretical (evaluated from a model) — its* barcode *is the ordered sequence of typed, frequency-tagged events:*

- ***Z*** *(zero-crossing): f such that $\gamma(f) = 0$ with a sign change.*
- ***M*** *(maximum): f such that γ has a local maximum.*
- ***N*** *(minimum): f such that γ has a local minimum.*

The name and visual framing come directly from how this looks when plotted: each event type rendered as a distinctly colored vertical mark along the frequency axis produces a pattern resembling a barcode. This is not merely a visualization convenience — it is the organizing metaphor for the whole method: a *good* spectral-estimation technique should produce a barcode that can be matched, stripe for stripe, against a barcode generated from physically plausible earth structure.

2.1 Template barcodes: analytically exact event locations

A *template* is the barcode of Equation 1 evaluated at one specific candidate phase-velocity curve $c(f)$. Because the theoretical curve is smooth and known in closed form (given $c(f)$), its barcode can be computed exactly, without any of the numerical sign-change-detection or derivative-estimation machinery needed for noisy real data.

This exactness follows from a classical Bessel-function identity used by Hawkins and Sambridge (2019, their Equations 1–3), extending the zero-crossing construction of Ekström et al. (2009, their Equation 2) to peaks and troughs as well:

- **Zero-crossings** of $J_0(x)$ occur at $x = j_{0,k}$, the k-th zero of J_0 (tabulated; in `scipy`, `scipy.special.jn_zeros(0, N)`).
- **Maxima and minima** of $J_0(x)$ occur where its derivative vanishes. Since $\frac{d}{dx}J_0(x) = -J_1(x)$, these occur at $x = j_{1,k}$, the k-th zero of J_1 (tabulated via `scipy.special.jn_zeros(1, N)`).

Given a candidate phase-velocity curve $c(f)$, the Bessel argument is $x(f) = \frac{2\pi f r}{c(f)}$. Template event frequencies are then obtained by solving

$$x(f_k) = j_{0,k} \quad \text{(Z events)}, \qquad x(f_k) = j_{1,k} \quad \text{(M/N events)} \tag{2}$$

for f_k. When $c(f)$ has no closed form (as is the case here, where $c(f)$ is built from a discretely sampled real dispersion curve — Section 4), this requires a one-dimensional numerical root-find (e.g., `scipy.optimize.brentq` against an interpolated $c(f)$) rather than a closed-form inversion. The key advantage over naive dense-sampling is that the *target* Bessel-argument value being solved for is exact and tabulated, not approximated by finite-difference sign-change or derivative detection on a sampled grid — there is no discretization error in identifying *which* x-values correspond to events, only ordinary root-finding tolerance in solving for the corresponding f.

Maxima and minima alternate strictly with each other and with zero-crossings in a fixed pattern (since J_0 and J_1's zeros interlace); this provides a useful internal consistency check when implementing the solver (Section 7).

2.2 Candidate barcodes: noise-contaminated, need reliability filtering

A *candidate* barcode is extracted from a real spectral estimate — FastMspec, single-taper (raw or smoothed), Mspec, or MspecBestK. Unlike templates, candidates have no closed form; their barcodes must be extracted numerically (sign-change detection for Z, local extrema detection for M/N) on the sampled, noisy estimate itself.

Real spectra contain spurious noise-driven events alongside genuine ones. Prior work in this project (Notebook 3, Section 1f) established that these cannot be separated by a single absolute threshold shared across techniques: a low-variance estimate (FastMspec) and a high-variance one (raw single-taper) have incomparable absolute amplitude-swing scales — a threshold calibrated from FastMspec's swing distribution would trivially pass 100% of single-taper's noise-driven events, since single-taper's noise alone produces larger local swings than FastMspec's entire signal.

The resolution, carried forward here: each method's events are filtered to "reliable" using that method's *own* relative threshold — e.g., the top 50% by local amplitude swing (maximum minus minimum in a small window straddling the event) within that method's own distribution, not an absolute value shared across techniques. This is applied uniformly to Z, M, and N events (turning-point prominence generalizes the same construction used for zero-crossing swing).

3 Matching and Scoring

Given a candidate barcode and one template barcode, a *match score* quantifies how well they agree. Four design elements, discussed in turn below: (1) a same-type, locally-scaled matching tolerance; (2) combined precision and recall, not recall alone; (3) hard rejection of physically-implausible matches, mirroring AkiNet's architectural (not merely penalized) bounding corridor; and (4) frequency-dependent trust weighting, following an explicit argument in Hawkins and Sambridge (2019) about where a reference earth model is most and least reliable.

3.1 Matching tolerance

A candidate event matches a template event of the same type if the two frequencies are within a tolerance. This tolerance is scaled to the *template's own local event spacing* at that point, not a fixed frequency value — consistent with the drift-as-a-percentage-of-local-spacing methodology already validated in Notebook 3 Section 1f. A fixed absolute tolerance would be too loose where template events are closely spaced (high frequency) and too tight where they are widely spaced (low frequency).

3.2 Precision and recall, not recall alone

A naive score — the fraction of template events that have a matching candidate event (recall) — is vulnerable to a specific failure mode: a very dense, noisy candidate can trivially "match" almost any template purely by having many more events than the template does. Concretely, in this project's own data, raw single-taper produces roughly 1,135 zero-crossings over a band where FastMspec produces 126; recall-only scoring would reward exactly the noise-dominated behavior the diagnostic is meant to penalize. The score therefore combines precision (fraction of candidate events that correspond to a real template event) and recall (fraction of template events that are matched) into a combined measure (e.g., their harmonic mean, analogous to an F1 score).

3.3 Hard corridor rejection

AkiNet (Xue and Olugboji, 2025, Section 3.3.1) bounds its inverted phase velocity to $c^{\mathrm{ivt}}(f) = W\hat{z}(f) + b(f)$, where $b(f)$ is a reference curve and W is a diagonal scaling matrix defining a fixed perturbation corridor (they use ± 0.8 km/s) around it. This is an *architectural* constraint: the network's output cannot leave the corridor regardless of what the data appear to want, not a soft penalty term traded off against data fit in a loss function.

This method mirrors that design choice directly, rather than relying only on soft precision/recall scoring: a candidate event whose *implied* local phase velocity (via the Ekström et al. 2009-style relation $c = \omega r/z$, evaluated at that event's own frequency and nearest Bessel-zero index) falls outside the *entire* template corridor — not merely "does not match the single best-fitting template," but "is not explainable by any corridor member at all" — is rejected as a candidate match, rather than merely counted as one more unmatched (but otherwise unremarkable) event. This distinguishes "noisy, but plausible" events from "implying an unphysical velocity outright," which a flat precision/recall score alone cannot.

3.4 Frequency-dependent trust

Hawkins and Sambridge (2019) argue directly for why a reference earth model should be trusted more at some frequencies than others: "most of the deviation between a prior Earth model and a new regional study will be confined to the near surface or higher frequency" (Discussion section). Long-period (low-frequency) surface waves sample deeper, larger-scale, more spatially averaged structure, which tends to vary less between a generic reference model and any specific region; short-period (high-frequency) waves sample shallow, heterogeneous structure where regional deviation from a reference model is expected and legitimate.

This method therefore weights mismatches (missed template events, spurious candidate events, or corridor-rejected events, Section 3.3) *more heavily* at the low-frequency end of the analysis band than at the high-frequency end. A technique that fails to reproduce the barcode structure where the reference model is most trustworthy is more damning than one that struggles only where legitimate regional deviation is expected. This replaces a naive frequency-uniform scoring scheme.

3.5 Diagnostic value of a null result

AkiNet's own discussion frames a specific failure mode — the network's output defaulting to the bare reference curve $b(f)$ (i.e., zero perturbation) — as a *useful, diagnostic* failure signature, not merely a bad outcome: it occurs when the input data are too poor to inform the fit at all, and its predictability makes such cases easy to flag for quality control (Xue and Olugboji, 2025, Section 4.4 and Discussion).

This method surfaces the analogous case explicitly: if a candidate's best-matching template turns out to be (or lies very close to) $\delta = 0$ (the bare reference curve, no perturbation), this is reported alongside the match score rather than silently absorbed into a single scalar. Combined with a poor absolute match score, it indicates the candidate carries too little real signal to usefully constrain anything, rather than being read as a confident "matches the reference well" result.

4 The Template Family

4.1 Reference curve: `SDISPL.ASC`

Templates require a base reference curve $c_{\text{ref}}(f)$ to build the perturbed family around (Section 4.2). A rough, hand-placed reference curve already existed elsewhere in this project (Notebook 3, Section 4, used for the `seislib` comparison), but its provenance was undocumented and it was explicitly flagged in this project's own notes as "NOT Sayan's exact curve."

A real, physically computed reference curve was located during this design process, on the lab network-attached storage (`repovibranium`), at `madagascar_data/pre_computed_files/SDISPL.ASC`. This is a standard Herrmann Computer-Programs-in-Seismology-format ASCII dispersion-curve file (columns: `LMODE`, `NFREQ`, `PERIOD(S)`, `FREQUENCY(Hz)`, `C(KM/S)`), containing a multi-mode Love-wave dispersion computation: the fundamental mode (`LMODE=0`) spans periods 2–2048 s (velocities 1.22–4.83 km/s), with higher modes (`LMODE=1--8`) each contributing starting at progressively shorter periods. This is the same file used by Sayan Swar's own `phasevel_compute_slide13and14.ipynb` (also on `repovibranium`, in `madagascar_codes/python/`) to build the reference curve shown in his project report's Figure 6, via exactly the same `LMODE==0` filter used here.

Restricted to this project's actual analysis band (0.02–0.3 Hz), the fundamental mode shows normal dispersion (velocity decreasing with increasing frequency): 4.49 km/s at 0.02 Hz down to 1.53 km/s at 0.3 Hz. This range brackets a much cruder empirical estimate derived independently elsewhere in this project (Notebook 3, Section 1f: $\sim$2.11 km/s from raw zero-crossing spacing) sensibly, which is itself a useful cross-check on both.

4.2 Building the template family

A single reference curve is not, by itself, "a range of plausible phase velocities." Following AkiNet's own design (Section 3.3), the template family is built as an *additive* perturbation corridor around the reference curve:

$$c_{\text{template}}(f) = c_{\text{ref}}(f) + \delta, \qquad \delta \in [-0.8, 0.8] \text{ km/s}, \tag{3}$$

stepped finely (e.g. 0.05 km/s steps, giving roughly 33 templates across the corridor). The corridor width, $\pm$0.8 km/s, is adopted directly from AkiNet's own tuned value — chosen by its authors, by experimentation, as "generous enough to span all plausible velocities required by the data" on the same problem domain (real Love-wave ambient noise) — rather than re-derived independently.

An earlier version of this design considered *multiplicative* scaling of the reference curve instead of additive perturbation; this was revised after reviewing AkiNet's own architecture, which is explicitly additive, not multiplicative.

4.3 The Love-wave Bessel-model simplification

Equation 1 (pure J_0) is, strictly, the Rayleigh-wave form. AkiNet (Xue and Olugboji, 2025, Equation 3, citing Nishida et al., 2024) gives the more physically correct Love-wave form as a combination of the zeroth- and second-order Bessel functions,

$$\rho(f, r) \approx \frac{1}{2}\left[J_0\left(\frac{2\pi f r}{c(f)}\right) - J_2\left(\frac{2\pi f r}{c(f)}\right)\right]. \tag{4}$$

This method uses pure J_0 for its first version, matching every prior use of the Bessel model elsewhere in this project (Notebook 2's illustrative example, `nb3_helpers.bessel_fit_ quality`, and Notebook 3 Section 1d's synthetic leakage test), and matching Hawkins and Sambridge's (2019) own practical simplification — their paper uses pure J_0 for both Love and Rayleigh data throughout, despite studying both wave types.

The trade-off is explicit: pure J_0 keeps the exact, tabulated-zero property that makes the analytical approach in Section 2.1 valuable in the first place. The combined $J_0 - J_2$ form does not have simply tabulated zeros (it is a difference of two different Bessel functions, not one), and would require numerical root-finding for template events too, losing most of the practical advantage of the analytical approach for this first version. This is recorded here as a known, deliberate simplification, not an oversight — a natural candidate for a future revision of this method.

5 Relationship to Existing Tools

This method is a genuine synthesis, not a restatement of any single existing tool in this project or in the literature reviewed while designing it:

- **`nb3_helpers.bessel_fit_quality`** (this project) grid-searches a constant-velocity trial c only (never a frequency-dependent $c(f)$), and fits the *full complex amplitude* of the observed coherence against the Bessel prediction via RMS residual. This project's own documentation already flags the resulting metric as unreliable, since it is "dominated by overall coherence strength, not Bessel-shape quality." A crossing/turning-point barcode is structurally immune to this specific failure mode: event *locations* do not depend on the overall amplitude scale the way a full RMS-amplitude fit does.

- **Notebook 3, Section 1f** (this project) checks only internal self-consistency of a candidate's own crossing spacing against itself; it never compares a candidate against an actual external physical template. This method's template library supplies exactly that missing external reference.

- **`seislib.extract_dispcurve`** (Magrini et al., 2022) is considerably more elaborate than this document originally characterized it, confirmed by a full read of its source (`_an_processing_seislib_functi` Sayan Swar's own copy, obtained from the same lab NAS location as the reference curve used here) rather than its high-level behavior alone. It is a four-stage, reference-curve-guided *sequential* zero-crossing tracker: (1) sign-change zero-crossing detection matched against tabulated J_0 zeros (or $J_0 - J_2$ zeros for the horizontal-polarization/Love-wave case, per Kästle et al., 2016, itself a direct methodological descendant of Ekström et al., 2009) to resolve candidate Bessel-branch indices; (2) a three-criterion `bad_quality` gate on each candidate crossing — adjacent-peak amplitude ratio exceeding 3.0, crossing amplitude below half a smoothed local envelope, or spacing to the previous crossing exceeding 1.5× the reference-predicted step; (3) elliptical-kernel density-field picking (weighted, KDE-like, around non-bad-quality crossings) with a local density-ratio acceptance threshold and cycle-jump-aware rejection of implausible velocity jumps; (4) a hard minimum-frequency-coverage acceptance test, returning a curve only if picks span enough of the band, else raising an exception with no partial result. **Critically, none of stages 2–4's internal quality signals are exposed to the caller** — the public return is binary (a curve, or an exception), and this project's `python/dispcurve_pick/` package vendors and lightly instruments exactly this file to surface those already-computed internals (bad-quality-crossing fraction, accepted-pick count, frequency-coverage fraction, mean pick amplitude ratio) as an explicit, continuous quality score, without altering the picking algorithm's own behavior. AkiNet's own paper (Xue and Olugboji, 2025) identifies "cycle skip" — committing early to the wrong phase oscillation — as this class of method's characteristic failure mode; seislib's own cycle-jump rejection and backup-picks mechanism (stage 3) is its internal defense against exactly that failure mode, not an omission this project needed to work around.

- **AkiNet** (Xue and Olugboji, 2025) solves the same underlying inverse problem via continuous physics-informed neural-network optimization rather than a discrete scan over templates. Its bounded-corridor architecture is philosophically identical to the template family used here (Section 4.2) — both encode "a plausible neighborhood around a physically motivated reference," arrived at independently via two very different mechanisms.

- **Ekström et al. (2009)** already constructs multiple trial phase-velocity curves from observed zero-crossings (indexed by an integer ambiguity m in their Equation 2) and selects whichever is closest to a reference curve — the same core "generate candidates, compare against a reference, pick the best" idea this method formalizes and extends, using an explicit multi-template precision/recall score (Section 3) rather than single-curve closest-match selection, and three event types instead of one.

- **Hawkins and Sambridge (2019)** extend that same zero-crossing construction to peaks and troughs via the J_1-zero identity (Section 2.1) — the direct analytical basis this method's template-event generation relies on — within a fundamentally different overall method (an adjoint-based nonlinear waveform-fitting inversion of a full path-average earth model, not a discrete barcode-matching scan).

The barcode/template-matching framing itself — three typed event streams, matched via combined precision, recall, corridor rejection, and frequency-weighted trust against a scanned template library — is original to this project, not a restatement of any of the above.

6 Scope and Open Items

- **Deliverable.** Both a reliability metric (for judging FastMspec against single-taper and other techniques) and a phase-velocity pick (the best-matching template's $c(f)$) — the latter a from-scratch alternative to `seislib.extract_dispcurve`, not a replacement for Notebook 3's existing Section 4, which remains as independent, unmodified work.
- **Data scope.** SA53/SA58 (Love-wave, transverse component) only for the first version. MTAN/RUNG — the pair where `seislib` previously failed to converge in this project, and the same pair AkiNet's own paper uses as a stress-test example — is an obvious and motivated next test, deferred to a later round once this method is validated on the cleaner pair.
- **Event weighting.** Z, M, and N event types are weighted equally in the match score for this first version; revisiting this (e.g. weighting crossings more heavily, since they tend to be more sharply localized than turning points) is a plausible future refinement.
- **Love-wave Bessel model.** Pure J_0 for this first version (Section 4.3); the more physically correct $J_0 - J_2$ combined form is a documented, deliberate simplification, not an oversight.

7 Implementation Notes

The companion notebook implementation should verify, before any results are trusted or reported:

1. `SDISPL.ASC` parses identically to Sayan Swar's own approach (`LMODE==0` filter, `[FREQUENCY_Hz, C_KM_S]` columns).

2. The template library is a sensible, non-degenerate fan of curves when plotted ($c_{\mathrm{ref}}(f)$ and several $c_{\mathrm{ref}}(f) + \delta$ members over 0.02–0.3 Hz).

3. The analytical event solver (Section 2.1) is correct: root-found frequencies should zero out $J_0(x(f))$ (or $J_1(x(f))$ for extrema) to near machine precision, and should agree with a brute-force dense-sampling/sign-change check on the same template curve; Z/M/N events should alternate in the expected order.

4. FastMspec's best-match score should exceed single-taper's (raw and roughness-matched smoothed variants) on real data — the expected result given everything established in Notebook 3 Sections 1c–1f. A result to the contrary should be treated as a bug to find, not a finding to report.

5. The picked best-fit velocity should be checked against both the raw reference curve's own value in the same band and the cruder empirical estimate from Section 1f, as a sanity range check.

8 Revision: From Barcode Matching to Instrumented Reference-Guided Picking

The scoring mechanism described in Sections 3–4.2 above — extracting a candidate's raw zero-crossing/maximum/minimum events by simple sign-change and neighbor-comparison detection, then scoring them via precision/recall against a scanned template library — was implemented and validated (Notebook 5, first version): it correctly ranked FastMspec above single-taper, and its own closing discussion honestly flagged a problem that turned out to be more than a minor caveat: *maximum/minimum detection produced roughly 5× as many "reliable" events as zero-crossing detection, dominated by per-frequency-bin noise wiggles rather than genuine curvature.* On reflection, and per direct guidance, this is not a tunable threshold problem — it is that naive local-event detection on raw noisy data is simply not a sound foundation for a quality metric, regardless of how the resulting events are later filtered or scored. Best practice, and the approach `seislib` itself takes, is to translate the coherence data into an actual phase-velocity curve *first*, using a picking process with real internal quality control, and to judge correlation quality from information available in that process — not to score raw spectral events directly.

This motivates the revision implemented from Notebook 5 v2 onward:

- **What is discarded.** The event-scanning scorer in its entirety — `candidate_barcode`, `match_score`, `_one_to_one_match`, `scan_templates`, and the raw Z/M/N neighbor-comparison extraction helpers. These remain reachable via git history (tag `notebook5-v1-event-scanning`) but are not carried forward.
- **What survives, repurposed.** The template family (Section 4.2: `load_reference_curve`, `build_template_family`, the `SDISPL.ASC` ± AkiNet-corridor construction) is unchanged, but its role changes: instead of being scanned as fixed targets for a candidate's raw events to match against, each member is fed as the `ref_curve` prior into seislib's own picker, and the best-performing prior (by convergence, coverage, and bad-quality fraction) is kept as the pick. This is the same "bounded neighborhood around a physically motivated reference" principle the original design shared with AkiNet (Section 5), now driving a real picker instead of a standalone scorer.
- **The new quality signal.** `python/dispcurve_pick/` vendors and instruments seislib's own `extract_dispcurve` (Section 5 above) to expose its already-computed internal diagnostics — bad-quality-crossing fraction, accepted-pick count, frequency-coverage fraction, mean pick amplitude ratio — as an explicit, continuous score, without altering the picking algorithm's own behavior (verified byte-identical to unmodified upstream when diagnostics are not requested). This directly answers what the original design could not: a real, physically grounded, already-battle-tested quality criterion, rather than one built from scratch on unreliable raw events.
- **Barcode visual, reframed.** The three-way Z/M/N coloring is replaced by seislib's own two-way good/bad-quality-crossing distinction — more honest given M/N was the noisiest, least trustworthy part of the original design, and directly tied to a real, principled criterion rather than an ad hoc relative-swing threshold.
- **Deferred, not abandoned: extending quality assessment to M/N.** The ambition behind a three-typed-event barcode was not wrong in principle — maxima and minima do carry real information (Hawkins and Sambridge, 2019) — only the naive detection-and-filtering implementation was. seislib's `bad_quality` criteria are defined specifically for zero-crossings; designing an analogous, principled quality gate for turning points remains a named, open item (tracked in `docs/notebook5_revamp_progress.md`), to be revisited once the zero-crossing-

based picker is validated at full dataset scale, not before.

- **Scale.** This revision is exercised across the full Madagascar dataset (380 station pairs, all four `ccf_pipeline` techniques: single-taper, FastMspec, Mspec, MspecBestK) processed on the university's HPC cluster, not the one-to-two-pair scale the original design was validated on — both because a real quality metric is only as convincing as the dataset it is shown to hold up across, and because this closes a gap `python/ccf_pipeline/NOTES.md` already named: the FastMspec path was, until this revision, verified only against synthetic data and Octave, never a real end-to-end SAC-to-output run at scale.

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